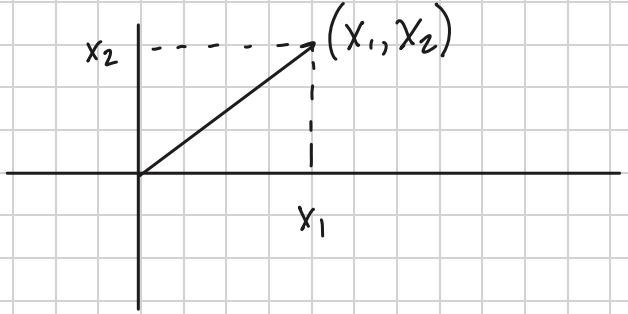
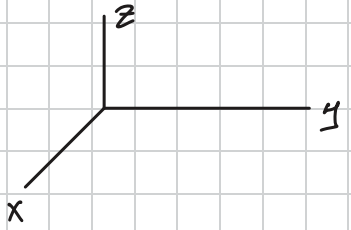
Week 7

$\mathbb{R}^2 \rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$





Standard bases

$\mathbb{R}^2$ Standard basis vectors

$$\begin{bmatrix}1\\0\end{bmatrix} \qquad \begin{bmatrix}0\\1\end{bmatrix}$$

e_1 e_2

Standard

Standard

$$\begin{bmatrix}x_1\\x_2\end{bmatrix} = x_1 \cdot \begin{bmatrix}1\\0\end{bmatrix} + x_2 \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}x_1\\0\end{bmatrix} + \begin{bmatrix}0\\x_2\end{bmatrix} = \begin{bmatrix}x_1\\x_2\end{bmatrix}$$

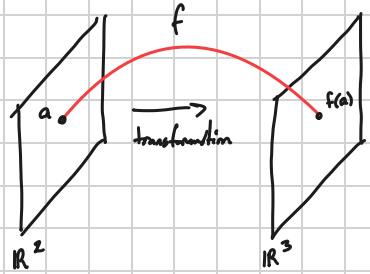
decompose or write in terms of linear comb of basis vectors

$$\begin{bmatrix}3\\1\end{bmatrix} = 3 \cdot e_1 + 1 \cdot e_2$$

Coefficients

$n=1$ $[x]$

+ define transformations/functions



$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \begin{matrix} -? \\ -? \\ -? \end{matrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Example $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} x_1 + 2x_2 \\ -x_2 \\ x_1^2 \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} 1 + 2 \cdot 0 \\ -0 \\ 1^2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

"how many transformations are there?"

matrix tranformations

T_A ← matrix that does the transfer

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \xrightarrow{T_A} \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = Y$$

$\mathbb{R}^n$ $\mathbb{R}^m$

$\dim(A) = m \times n$

$$\overline{X} \xrightarrow{T_A} \overline{Y} = A \cdot \overline{X}$$

matrix transformation

$\mathbb{R}^n \xrightarrow{T_A} \mathbb{R}^m$ is a matrix multiplication

Checks

$\dim(Y) = m \times 1$ (output)

$\dim(X) = n \times 1$ (input)

$$\overline{Y} = A \cdot \overline{X}$$

$$(m \times n) \cdot (n \cdot 1) = (m \times 1)$$

$\dim(A)$ $\dim(X)$

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 0 \\ 4 & 3 \end{bmatrix}$$

$\dim(A) = 3 \times 2$

$\quad\quad\quad

Matrix transformations are linear (will be on midterm)

rules to be satisfied

① $T_A(k \cdot \vec{x}) = k \cdot T_A(\vec{x})$

② $T_A(X + Y) = T_A(X) + T_A(Y)$

Ex

$\begin{bmatrix} 3 \\ 1 \end{bmatrix} \xrightarrow{T_A} \begin{bmatrix} \pi \\ -\frac{1}{2} \\ 7 \end{bmatrix}$

if you know its is linear

② $\cdot \begin{bmatrix} 3 \\ 1 \end{bmatrix} \rightarrow 2 \begin{bmatrix} \pi \\ -\frac{1}{e} \\ 7 \end{bmatrix} = \begin{bmatrix} 2\pi \\ -\frac{2}{e} \\ 14 \end{bmatrix}$

$\begin{bmatrix} 2 \\ 6 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \xrightarrow{T_A} \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$

$\begin{bmatrix} -1 \\ 1 \end{bmatrix} \xrightarrow{T_A} \begin{bmatrix} 0 \\ 6 \\ 7 \end{bmatrix}$

$\begin{bmatrix} 0 \\ 2 \end{bmatrix} \xrightarrow{?}$

$\begin{bmatrix} 0 \\ 2 \end{bmatrix} \quad 2T \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ Die

Any matrix transformation is linear

Examples for midterm

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \longrightarrow \begin{bmatrix} x_1 + 2x_2 \\ 0 \end{bmatrix}$$

$\mathbb{R}^2$ $\qquad$ $\mathbb{R}^2$

Proof (it is linear)

- $\begin{bmatrix} k \cdot x_1 \\ k \cdot x_2 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} kx_1 + 2(kx_2) \\ 0 \end{bmatrix} = k \cdot \begin{bmatrix} x_1 + 2x_2 \\ 0 \end{bmatrix}$ (factor out)

- $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \xrightarrow{T}$

$\downarrow$

$$\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} (x_1 + y_1) + 2(x_2 + y_2) \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 + 2x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} y_1 + 2y_2 \\ 0 \end{bmatrix}$$

$$T\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad T\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Matrix transformations:

$$\mathbb{R}^n \xrightarrow{T_A} \mathbb{R}^m$$

$$\bar{x} \in \mathbb{R}^n : \quad \bar{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \Big\} n\text{-entries}$$

$$\bar{y} \in \mathbb{R}^m : \quad \bar{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} \Big\} m\text{-entries}$$

$$\dim(A) = m \times n$$

$$\bar{x} \xrightarrow{T_A} \bar{y} = A \cdot \bar{x}$$

> Any linear transformation T is a matrix transformation

if "T" is linear → should be able to find "A"

$$\mathbb{R}^3 \xrightarrow{T} \mathbb{R}^4 : \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} x_1 - 2x_3 \\ -40x_1 + x_2 \\ x_1 - \pi x_2 + 10x_3 \\ 4x_2 + 100x_3 \end{bmatrix}$$

$$\overline{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad \overline{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

Checks that T is "linear"

$$\dim(A) = m \times n = 4 \times 3$$

↳ Standard basis: ($\mathbb{R}^n$ in space)

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \qquad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Always

$$A = \underbrace{\begin{bmatrix} T(e_1) & T(e_2) & T(e_3) \end{bmatrix}}_{n=3 \text{ colum}}$$

$m=4$ raw

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} 1 \\ -40 \\ 1 \\ 0 \end{bmatrix}$$

e_1 $\qquad$ $T(e_1)$

$x_1 = 0$
$x_2 = 0$
$x_3 = 0$

$$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} 0 \\ 1 \\ -\pi \\ 4 \end{bmatrix}$$

$T(e_2)$

$$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} -2 \\ 0 \\ 10 \\ 100 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & -2 \\ -40 & 1 & 0 \\ 1 & -\pi & 10 \\ 0 & 4 & 100 \end{bmatrix}$$

Holy Slow ff Down

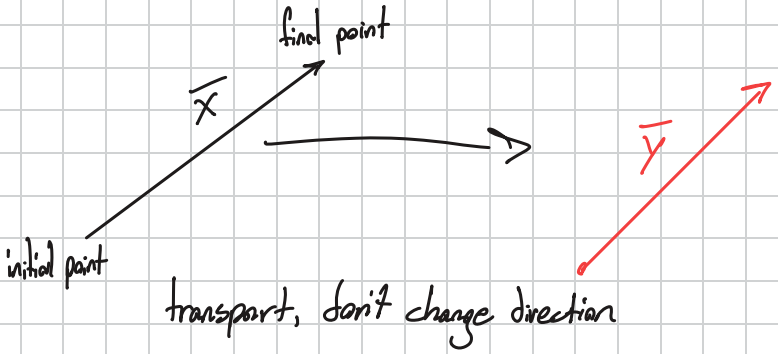
Lecture 16 (we skiped something thats on the test)

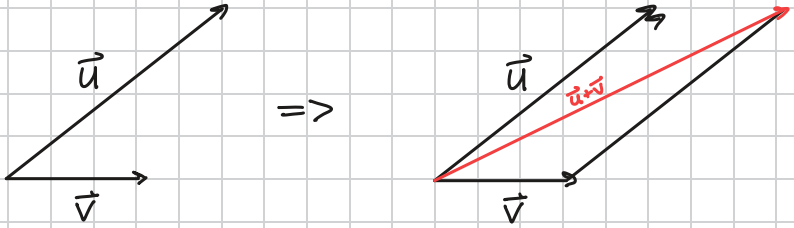
$\mathbb{R}^2$: $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Word of the day Euclidean

tells you how to compute:

distance, length, angles





$0 \cdot \vec{x} = \vec{0}$ $\quad$ $\vec{x} + \vec{0} = \vec{x}$

In general for $n \geq 3$ (algebraically)

Linear combination

=> you need same number of vectors (in some $\mathbb{R}^n$)

$$\begin{bmatrix}1\\2\end{bmatrix} + \begin{bmatrix}1\\2\\3\end{bmatrix} = \;\rightarrow \text{undefined}$$

$\rightarrow$ Python: ✗

python cheats we no like

$$\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}-1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}$$

$\vec{u}_1, u_2, u_3$

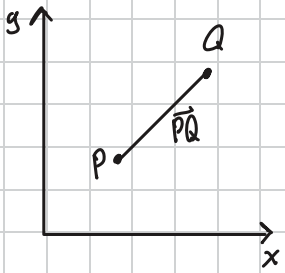
=> you need three numbers

$c_1 = 1$, $c_2 = \sqrt{2}$, $c_3 = \pi$

=> linear combination is:

$$c_1\vec{u}_1 + c_2\vec{u}_2 + c_3\cdot\vec{u}_3$$

$= 1\begin{bmatrix}1\\2\end{bmatrix} + \sqrt{2}\begin{bmatrix}-1\\0\end{bmatrix} + \pi\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}1-\sqrt{2}\\2+\pi\end{bmatrix}$



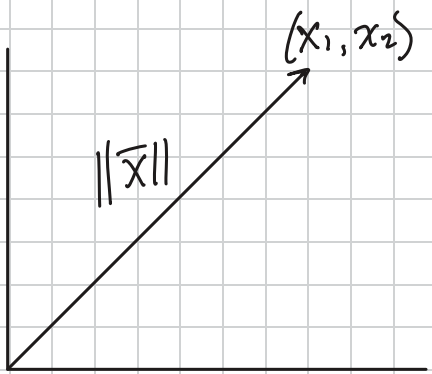
then

$$\overrightarrow{PQ} \equiv (x_2 - x_1,\ y_2 - y_1)$$

$$P = \begin{bmatrix} x_P \\ y_P \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \overrightarrow{PQ} \overset{\text{def}}{=} \begin{bmatrix} 3 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$Q = \begin{bmatrix} x_Q \\ y_Q \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

final initial



distance (P, Q)

$(P, Q) \longrightarrow \overrightarrow{PQ}$ $\quad dist(P,Q) = |\overrightarrow{PQ}|$

$$d(P,Q) = \sqrt{(x_Q - x_P)^2 + (y_Q - y_P)^2}$$

only one vector $|\vec{v}| = 0$

$\hookrightarrow \vec{v}^2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\|\vec{v}\| = 0$

$\mathbb{R}^4$: